INSTRUMENT CLUSTER INSPECTION

id092200010600

Speedometer Inspection (Analog Speedometer Type)
Using a speedometer tester

Caution

- **When only the front or the rear wheels are driven using a speedometer tester, the DSC HU/CM may determine that the vehicle speed signal is in error and the engine cannot be stopped by the normal operation. If the engine cannot be stopped, perform an emergency engine stop operation. (See SERVICE CAUTIONS.)**

1. Adjust the tire pressure to the specification. (See WHEEL AND TIRE SPECIFICATION.)
2. Using a speedometer tester, verify that the values are as indicated in the table.

European (L.H.D. U.K.), Australian specs.

Speedometer indication (km/h)	Speedometer tester indication (km/h)
20	15.9—19.9
40	35.4—39.4
60	55—59
80	74.5—78.5
100	94.1—98.1
120	112.25—117.75
140	131.4—137.4
160	151—157
180	170.6—176.6
200	190.2—196.2

General (L.H.D. R.H.D.) specs.

Speedometer indication (km/h)	Speedometer tester indication (km/h)
20	18.0—22.0
40	38.0—42.0
60	58.0—62.0
80	78.0—82.0
100	98.0—102.0
120	117.25—122.75
140	137.0—143.0
160	157.0—163.0
180	177.0—183.0
200	197.0—203.0

U.K. specs.

Speedometer indication (mph)	Speedometer tester indication (mph)
10	8.748—11.248
20	18.75—21.25
30	28.65—31.15
40	38.74—41.24
50	48.74—51.24
60	58.74—61.24
70	68.485—71.485
80	78.233—81.733
90	88.181—91.781
100	98.129—101.829
110	108.13—111.83
120	118.12—121.82
130	127.82—132.12

- If not as indicated in the table, perform the DTC inspection and related wiring harness inspection for the following modules.
  - — PCM
  - — DSC HU/CM

- If the PCM, DSC HU/CM, and related wiring harnesses are normal, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

Using the M-MDS

1. Connect the M-MDS to the DLC-2.
2. After vehicle identification, select the following from the M-MDS initialization screen.
   (1) "Data logger"
   (2) "Module"
   (3) "IC"
3. Using the simulation function SPEEDOMTR, verify that the values are as indicated in the table.

European (L.H.D. U.K.), Australian specs.

M-MDS display	Speedometer display
60 km/h	Speedometer needle moves to approx. 61—65 km/h {38—40 mph}.
120 km/h	Speedometer needle moves to approx. 122-128 km/h {75.9—79.5 mph}.
Off	Speedometer needle moves to the bottommost position.

General (L.H.D. R.H.D.) specs.

M-MDS display	Speedometer display
60 km/h	Speedometer needle moves to approx. 58—62 km/h {37—38 mph}.
120 km/h	Speedometer needle moves to approx. 117-123 km/h {72.8—76.4 mph}.
Off	Speedometer needle moves to the bottommost position.

- If not as indicated in the table, perform the related wiring harness inspection.
  — If the related wiring harnesses are normal, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

Speedometer Inspection (Digital Speedometer Type)

Caution

- **When only the front or the rear wheels are driven using a speedometer tester, the DSC HU/CM may determine that the vehicle speed signal is in error and the engine cannot be stopped by the normal operation. If the engine cannot be stopped, perform an emergency engine stop operation. (See SERVICE CAUTIONS.)**

1. Adjust the tire pressure to the specification. (See WHEEL AND TIRE SPECIFICATION.)
2. Using a speedometer tester, verify that the values are as indicated in the table.

European (L.H.D. U.K.), General (R.H.D.), Australian specs.

Speedometer indication (km/h)	Speedometer tester indication (km/h)
20	20.4—22.4
40	40.8—42.8
60	61.2—63.2
80	81.6—83.6
100	102—104
120	122.4—124.4
140	142.8—144.8
160	163.2—165.2
180	183.6—185.6
200	204—206
220	224.4—226.4
240	244.8—246.8

General (L.H.D.) specs.

Speedometer indication (km/h)	Speedometer tester indication (km/h)
20	19.0—21.0
40	39.0—41.0
60	59.0—61.0
80	79.0—81.0
100	99.0—101.0
120	119.0—121.0
140	139.0—141.0
160	159.0—161.0
180	179.0—181.0
200	199.0—201.0
220	219.0—221.0
240	239.0—241.0

- If not as indicated in the table, perform the DTC inspection and related wiring harness inspection for the following modules.
  - — PCM
  - — DSC HU/CM
- If the PCM, DSC HU/CM, and related wiring harnesses are normal, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

Tachometer Inspection (Analog Tachometer type)

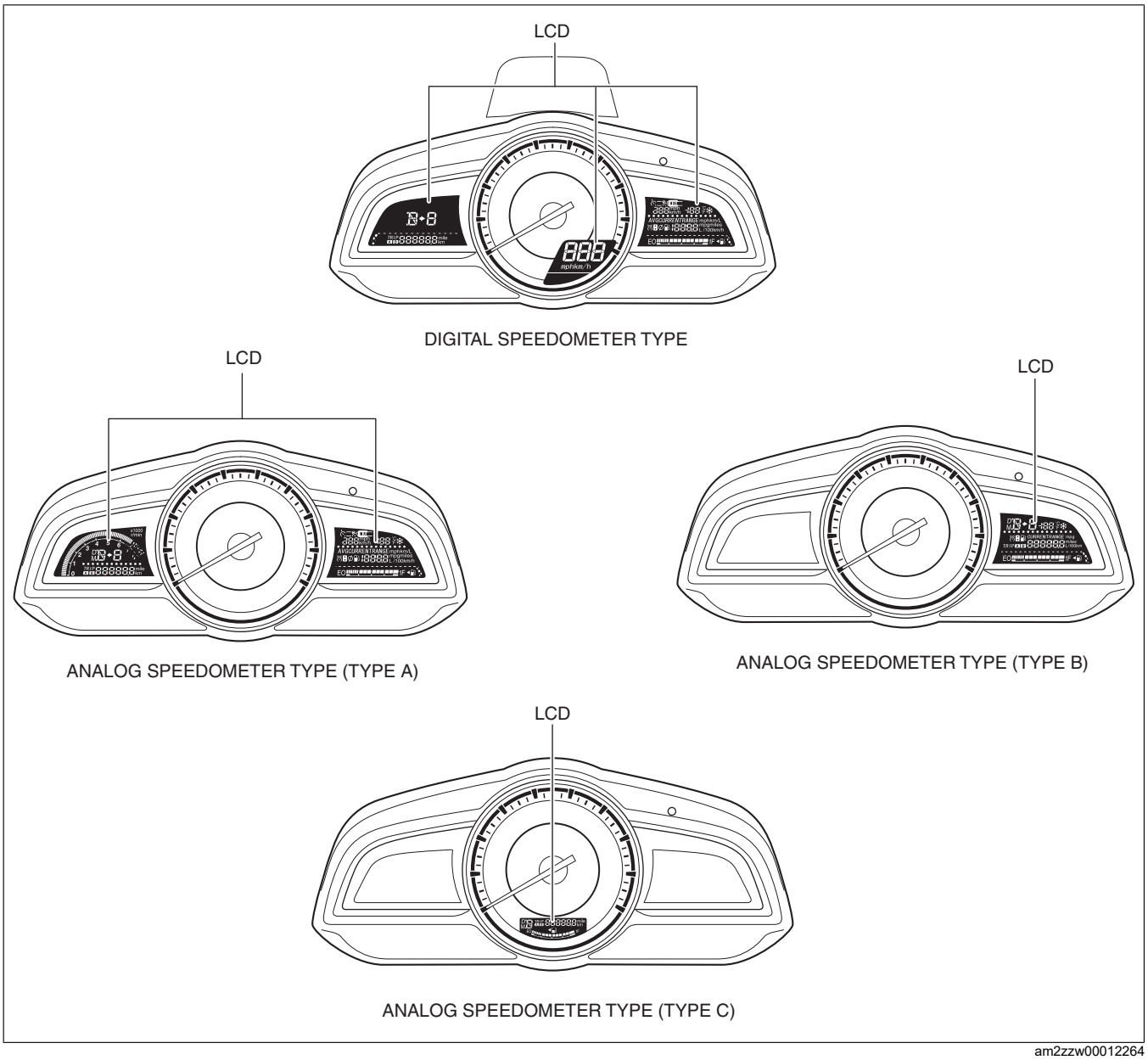
Caution
- **If the engine speed exceeds the allowable range, the engine could be damaged. Therefore, when inspecting the tachometer, do not allow the engine speed to exceed the allowable range indicated on the tachometer.**

1. Connect the M-MDS to the DLC-2.
2. After vehicle identification, select the following from the M-MDS initialization screen.
   (1) “Data logger”
   (2) “Module”
   (3) “IC”
3. Using the simulation function TACHOMTR, verify that the values are as indicated in the table.

M-MDS display	Tachometer display
3000 RPM	Tachometer needle moves to approx. 3,050 rpm.
6000 RPM	Tachometer needle moves to approx. 6,100 rpm.
Off	Tachometer needle moves to bottommost position.

- If not as indicated in the table, perform the related wiring harness inspection.
  - — If the related wiring harnesses are normal, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

LCD Inspection
1. Connect the M-MDS to the DLC-2.
2. After vehicle identification, select the following from the M-MDS initialization screen.
   (1) “Data logger”
   (2) “Module”
   (3) “IC”
3. Using the simulation function LCD_SEG, verify that the LCD is illuminated.
   - If there are any non-illuminated area, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

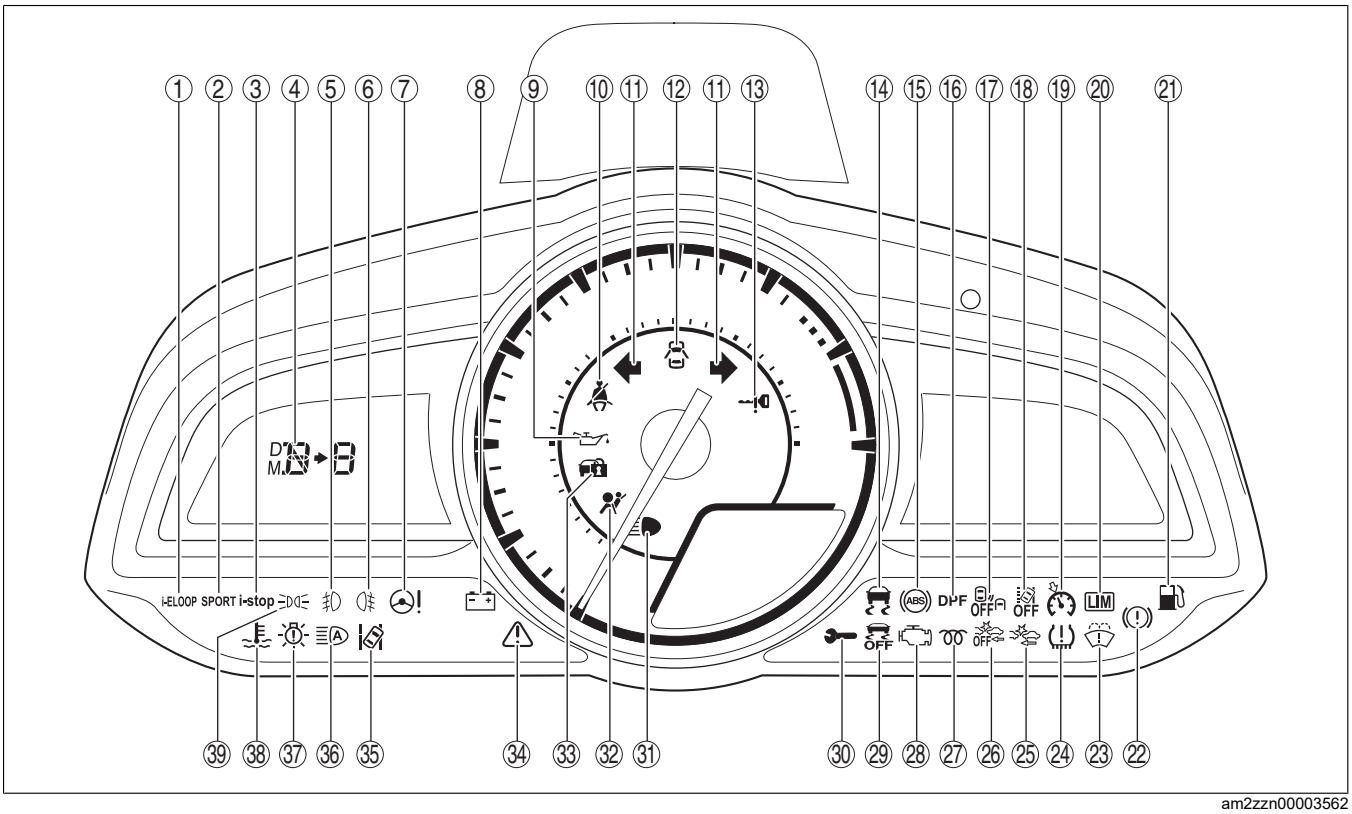


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Warning/Indicator Light Illumination Inspection

1. Connect the M-MDS to the DLC-2.
2. After vehicle identification, select the following from the M-MDS initialization screen.
   (1) “Data logger”
   (2) “Module”
   (3) “IC”
3. Using the simulation function WL+IL, verify that the warning/indicator lights in the following table are illuminated.
   - If any of the warning/indicator lights are not displayed, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

Digital speedometer type



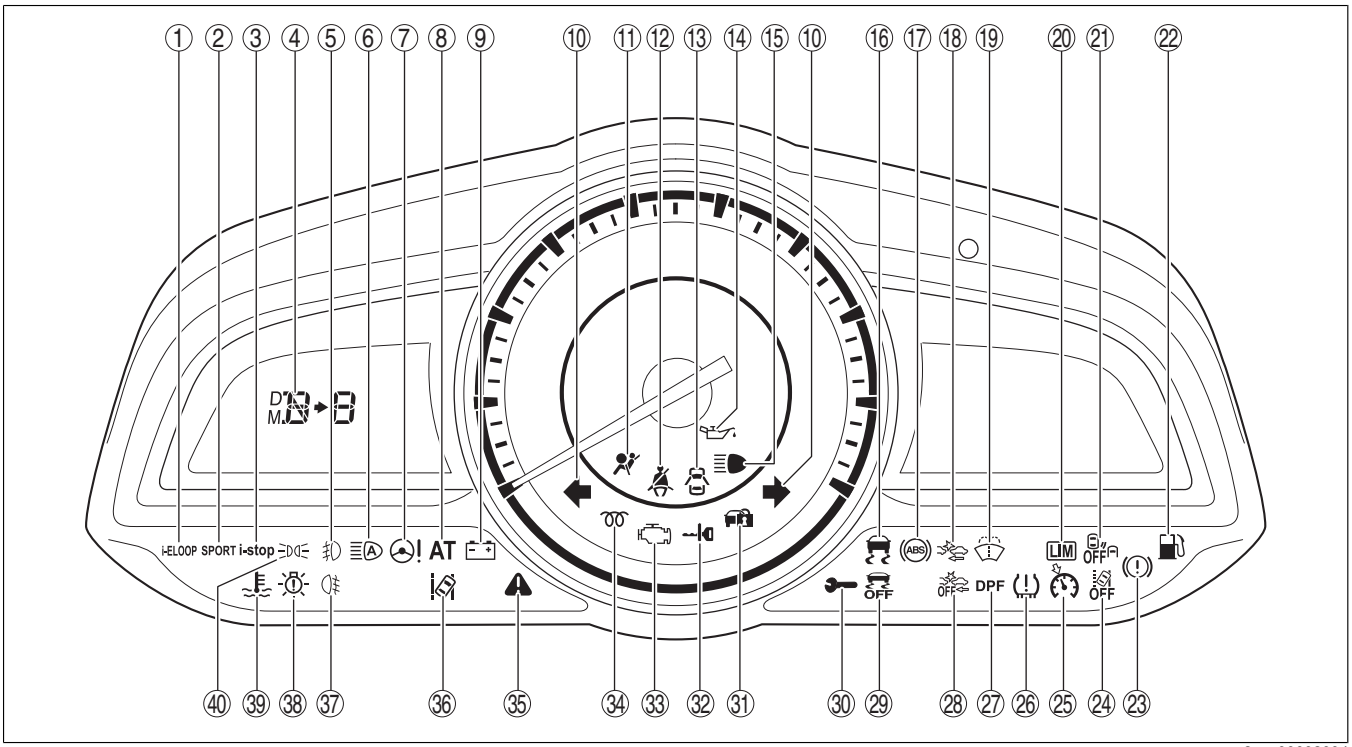
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X: Applicable
—: Not applicable

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
1	i-ELOOP	i-ELOOP indicator light (green)/i-ELOOP warning light (amber)	—	With i-ELOOP	—
2	SPORT	Select mode indicator light	—	SKYACTIV-G 1.3, SKYACTIV-G 1.5	—
3	i-stop	i-stop warning light (amber)/i-stop indicator light (green)	×	With i-stop system	×
4		Gear shift indicator	—	—	—
5		Fog light indicator light	—	With front fog lights	—
6		Rear fog light indicator light	—	With rear fog light	—
7		Power steering malfunction indicator light	×	—	×
8		Charging system warning light	×	—	×
9		Engine oil warning light	×	—	×
10		Seat belt warning light	—	—	—
11		Turn signal/hazard warning indicator lights	—	—	—
12		Door-ajar warning light	—	—	—
13		KEY warning light (red)/KEY indicator light (green)	×	—	×
14		TCS/DSC indicator light	×	—	×
15	ABS	ABS warning light	×	—	×
16	DPF	Diesel particulate filter indicator light	×	SKYACTIV-D 1.5	×

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
17		Blind spot monitoring (BSM) OFF indicator light	—	With blind spot monitoring (BSM) system	×
18		LDWS OFF indicator light	—	With lane departure warning system (LDWS)	×
19		Cruise main indicator light (amber)/cruise set indicator light (green)	—	With cruise control system	—
20		Adjustable speed limiter main indicator light (amber)/adjustable speed limiter set indicator light (green)	—	With adjustable speed limiter	—
21		Low fuel warning light	×	—	—
22		Brake system warning light	×	—	×
23		Low washer fluid level warning light	×	With washer fluid-level sensor	—
24		Tire pressure monitoring system warning light	×	With TPMS	×
25		Smart city brake support (SCBS) indicator light (red)/smart city brake support (SCBS) warning light (amber)	—	With smart city brake support (SCBS)	×
26		Smart city brake support (SCBS) OFF indicator light	—	With smart city brake support (SCBS)	×
27		Glow indicator light	×	SKYACTIV-D 1.5	×
28		Check engine light	×	—	×
29		DSC OFF indicator light	—	—	×
30		Wrench indicator light	—	—	×
31		Headlight high-beam indicator light	—	—	—
32		Air bag/seat belt pretensioner system warning light	×	—	×
33		Security indicator light	—	—	×
34		Master warning light	×	—	×
35		LDWS warning light	×	With lane departure warning system (LDWS)	×
36		HBC indicator light (green)/HBC warning light (amber)	×	With high beam control (HBC) system	×
37		LED headlight warning light	—	With front combination light (LED type)	×
38		Low engine coolant temperature indicator light (blue)/high engine coolant temperature warning light (red)	×	—	×
39		Lights-on indicator light	—	—	—

Analog speedometer type (Type A)



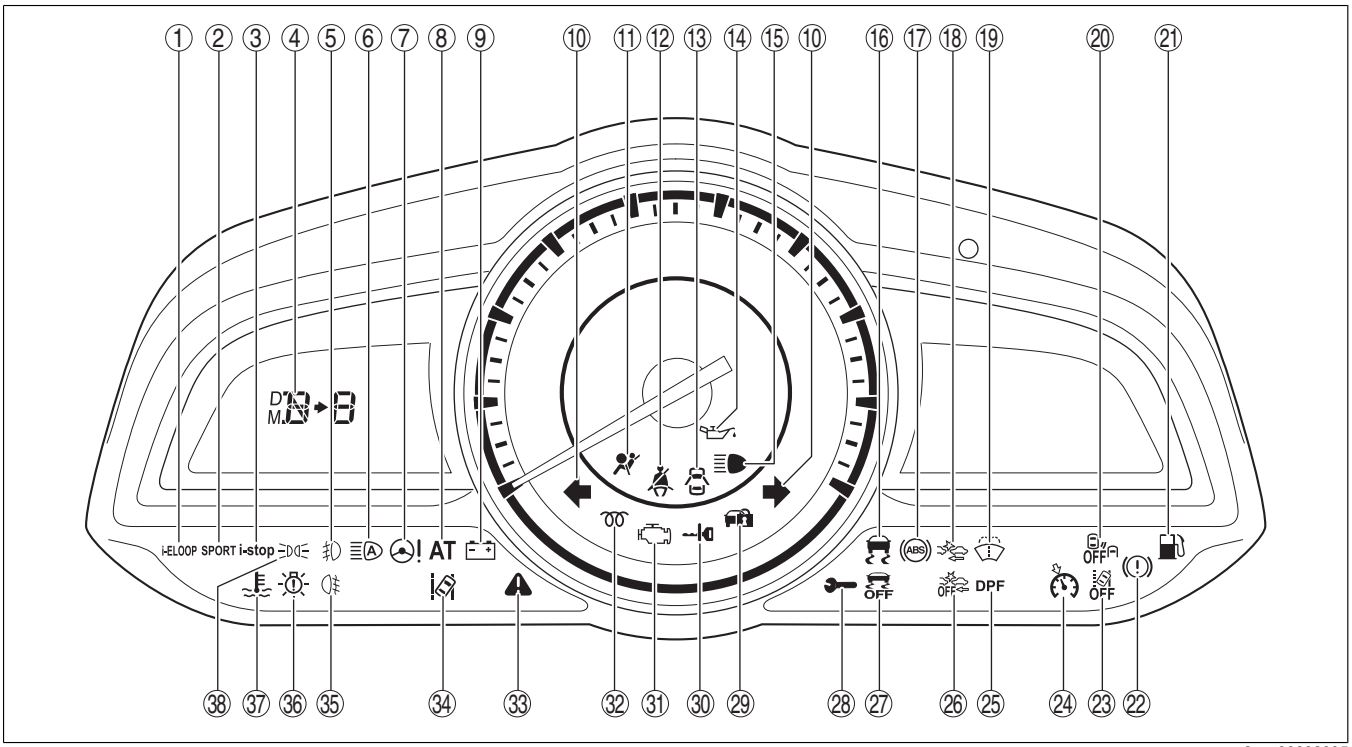
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X: Applicable
—: Not applicable

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
1	i-ELOOP	i-ELOOP indicator light (green)/i-ELOOP warning light (amber)	—	With i-ELOOP	—
2	SPORT	Select mode indicator light	—	SKYACTIV-G 1.3, SKYACTIV-G 1.5 (MTX)	—
3	i-stop	i-stop warning light (amber)/i-stop indicator light (green)	×	With i-stop system	×
4	D M 6 → 8	Shift position indicator/gear position indicator/gear shift indicator	—	—	—
5		Fog light indicator light	—	With front fog lights	—
6	A	HBC indicator light (green)/HBC warning light (amber)	×	With high beam control (HBC) system	×
7	!	Power steering malfunction indicator light	×	—	×
8	AT	Automatic transaxle warning light	×	ATX	×
9		Charging system warning light	×	—	×
10		Turn signal/hazard warning indicator lights	—	—	—
11		Air bag/seat belt pretensioner system warning light	×	—	×
12		Seat belt warning light	—	—	—
13		Door-ajar warning light	—	—	—
14		Engine oil warning light	×	—	×
15		Headlight high-beam indicator light	—	—	—
16		TCS/DSC indicator light	×	—	×

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
17		ABS warning light	×	—	×
18		Smart city brake support (SCBS) indicator light (red)/smart city brake support (SCBS) warning light (amber)	—	With smart city brake support (SCBS)	×
19		Low washer fluid level warning light	×	With washer fluid-level sensor	—
20		Adjustable speed limiter main indicator light (amber)/adjustable speed limiter set indicator light (green)	—	With adjustable speed limiter	—
21		Blind spot monitoring (BSM) OFF indicator light	—	With blind spot monitoring (BSM) system	×
22		Low fuel warning light	×	—	—
23		Brake system warning light	×	—	×
24		LDWS OFF indicator light	—	With lane departure warning system (LDWS)	×
25		Cruise main indicator light (amber)/cruise set indicator light (green)	—	With cruise control system	—
26		Tire pressure monitoring system warning light	×	With TPMS	×
27	DPF	Diesel particulate filter indicator light	×	SKYACTIV-D 1.5	×
28		Smart city brake support (SCBS) OFF indicator light	—	With smart city brake support (SCBS)	×
29		DSC OFF indicator light	—	—	×
30		Wrench indicator light	—	—	×
31		Security indicator light	—	—	×
32		KEY warning light (red)/KEY indicator light (green)	×	—	×
33		Check engine light	×	—	×
34		Glow indicator light	×	SKYACTIV-D 1.5	×
35		Master warning light	×	—	×
36		LDWS warning light	×	With lane departure warning system (LDWS)	×
37		Rear fog light indicator light	—	With rear fog light	—
38		LED headlight warning light	—	With front combination light (LED type)	×
39		Low engine coolant temperature indicator light (blue)/high engine coolant temperature warning light (red)	×	—	×
40		Lights-on indicator light	—	—	—

Analog speedometer type (Type B)



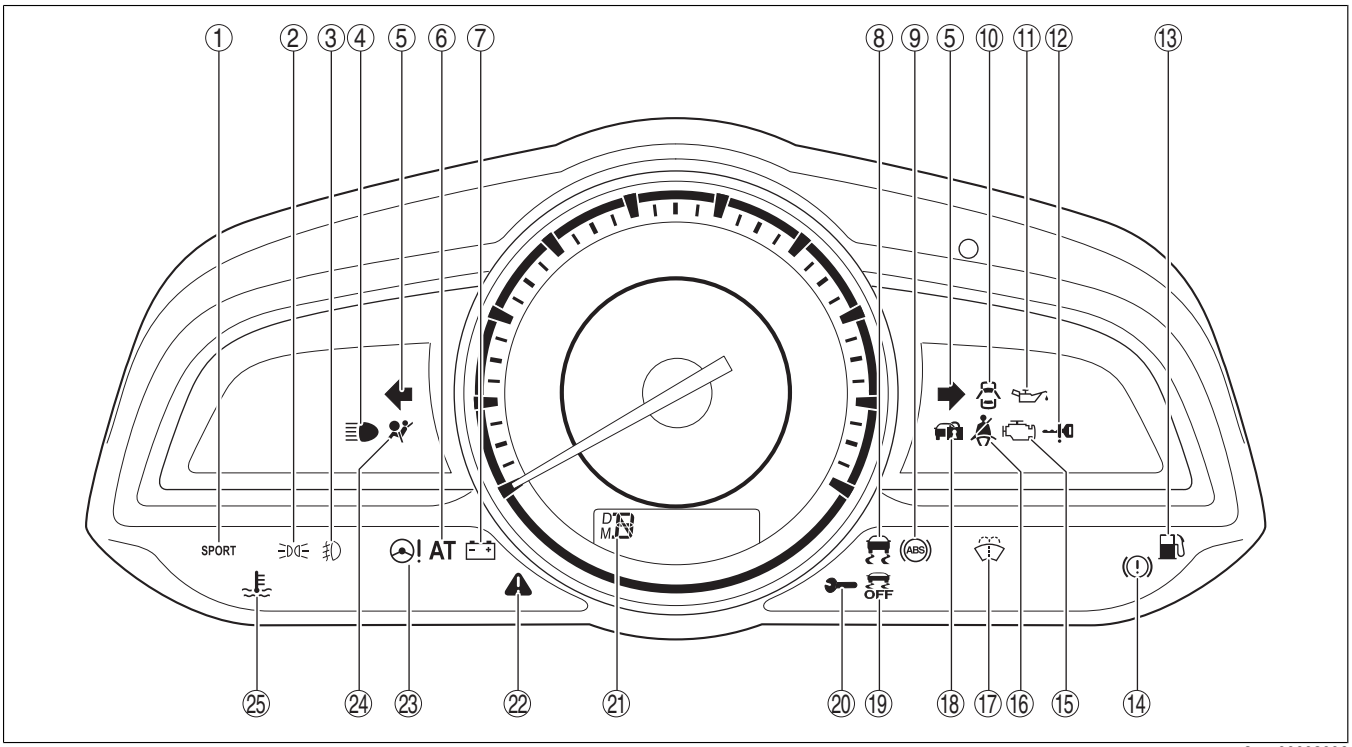
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X: Applicable
—: Not applicable

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
1	i-ELOOP	i-ELOOP indicator light (green)/i-ELOOP warning light (amber)	—	With i-ELOOP	—
2	SPORT	Select mode indicator light	—	SKYACTIV-G 1.3, SKYACTIV-G 1.5 (MTX)	—
3	i-stop	i-stop warning light (amber)/i-stop indicator light (green)	×	With i-stop system	×
4	D M 6 → 8	Shift position indicator/gear position indicator/gear shift indicator	—	—	—
5		Fog light indicator light	—	With front fog lights	—
6	A	HBC indicator light (green)/HBC warning light (amber)	×	With high beam control (HBC) system	×
7	!	Power steering malfunction indicator light	×	—	×
8	AT	Automatic transaxle warning light	×	ATX	×
9		Charging system warning light	×	—	×
10		Turn signal/hazard warning indicator lights	—	—	—
11		Air bag/seat belt pretensioner system warning light	×	—	×
12		Seat belt warning light	—	—	—
13		Door-ajar warning light	—	—	—
14		Engine oil warning light	×	—	×
15		Headlight high-beam indicator light	—	—	—
16		TCS/DSC indicator light	×	—	×

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
17		ABS warning light	×	—	×
18		Smart city brake support (SCBS) indicator light (red)/smart city brake support (SCBS) warning light (amber)	—	With smart city brake support (SCBS)	×
19		Low washer fluid level warning light	×	With washer fluid-level sensor	—
20		Blind spot monitoring (BSM) OFF indicator light	—	With blind spot monitoring (BSM) system	×
21		Low fuel warning light	×	—	—
22		Brake system warning light	×	—	×
23		LDWS OFF indicator light	—	With lane departure warning system (LDWS)	×
24		Cruise main indicator light (amber)/cruise set indicator light (green)	—	With cruise control system	—
25	DPF	Diesel particulate filter indicator light	×	SKYACTIV-D 1.5	×
26		Smart city brake support (SCBS) OFF indicator light	—	With smart city brake support (SCBS)	×
27		DSC OFF indicator light	—	—	×
28		Wrench indicator light	—	—	×
29		Security indicator light	—	—	×
30		KEY warning light (red)/KEY indicator light (green)	×	—	×
31		Check engine light	×	—	×
32		Glow indicator light	×	SKYACTIV-D 1.5	×
33		Master warning light	×	—	×
34		LDWS warning light	×	With lane departure warning system (LDWS)	×
35		Rear fog light indicator light	—	With rear fog light	—
36		LED headlight warning light	—	With front combination light (LED type)	×
37		Low engine coolant temperature indicator light (blue)/high engine coolant temperature warning light (red)	×	—	×
38		Lights-on indicator light	—	—	—

Analog speedometer type (Type C)



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X: Applicable
—: Not applicable

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
1	SPORT	Select mode indicator light	—	MTX	—
2		Lights-on indicator light	—	—	—
3		Fog light indicator light	—	With front fog lights	—
4		Headlight high-beam indicator light	—	—	—
5		Turn signal/hazard warning indicator lights	—	—	—
6	AT	Automatic transaxle warning light	×	ATX	×
7		Charging system warning light	×	—	×
8		TCS/DSC indicator light	×	—	×
9	ABS	ABS warning light	×	—	×
10		Door-ajar warning light	—	—	—
11		Engine oil warning light	×	—	×
12		KEY warning light (red)/KEY indicator light (green)	×	—	×
13		Low fuel warning light	×	—	—
14		Brake system warning light	×	—	×
15		Check engine light	×	—	×
16		Seat belt warning light	—	—	—
17		Low washer fluid level warning light	×	With washer fluid-level sensor	—
18		Security indicator light	—	—	×
19	OFF	DSC OFF indicator light	—	—	×
20		Wrench indicator light	—	—	×

No	Warning light/ Indicator light	Name	Message is displayed in center display when warning/indicator light turns on/ flashes	Comment	Turns on when IG-ON
21		Shift position indicator/gear position indicator/gear shift indicator	—	—	—
22		Master warning light	×	—	×
23		Power steering malfunction indicator light	×	—	×
24		Air bag/seat belt pretensioner system warning light	×	—	×
25		Low engine coolant temperature indicator light (blue)/high engine coolant temperature warning light (red)	×	—	×

Alarm Inspection

1. Connect the M-MDS to the DLC-2.
2. After vehicle identification, select the following from the M-MDS initialization screen.
   (1) “Data logger”
   (2) “Module”
   (3) “IC”
3. Using the simulation function ALARM, verify that the buzzer sounds.
   - If the buzzer does not sound, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)